Successive Convex Optimization, Facility Location, UAV Path Planning and an SDP

Convex Optimization, Comprehensive Homework

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English summary of my original Persian report, which is kept as a PDF in this folder.

1. Successive convex optimization

This is a local method for non-convex problems. Each step replaces the non-convex parts with convex approximations around the current point (linearisation or convex upper bounds), solves the convex problem, and repeats. It usually converges to a local optimum. A global optimum is guaranteed only under special structure, such as difference-of-convex problems with extra conditions.

2. Warehouse location as set covering

For demand points s_i, candidate sites d_j with build costs $c_j > 0$, range R and at most k warehouses, I let $y_j \in \{0, 1\}$ mean "build at d_j" and defined $N_i = \{j : \|d_j - s_i\| \le R\}$. The model is

$$\min_{y} \sum_{j} c_j y_j \quad \text{s.t.} \quad \sum_{j \in N_i} y_j \ge 1 \;\; \forall i, \qquad \sum_{j} y_j \le k.$$

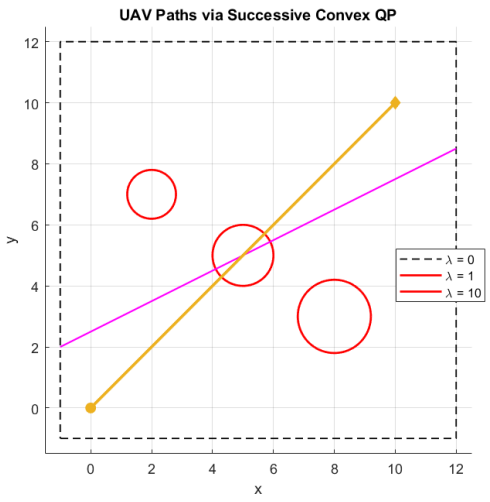
3. UAV path planning (`Q3.m`)

The cost is

$$J = \sum_{t=0}^{T-1} \left(\|u_t\|^2 + \alpha \|x_{t+1} - x_t\|^2 \right) + \lambda \, R(x_0, \ldots, x_T).$$

It has double-integrator dynamics, box limits on velocity and acceleration, fixed start and goal, three circular obstacles and two linear danger zones. I compared hard obstacle constraints with a penalty approach. The circle constraints are linearised around the current path, each zone gets a slack variable penalised by λs^2, and every iteration solves a QP with `quadprog` (linearise, solve, update with damping). The report runs three cases: $\lambda = 0$ gives the straight, risky path, $\lambda = 1$ a balanced one, and $\lambda = 10$ a conservative one.

Re-running the script: `quadprog` reports the linearised QP as infeasible at every iteration (exit flag -2), so the script falls back to its initial straight-line guess and all three curves coincide. I left the code as submitted.



4. A small SDP (`Q4.m`, CVX)

The problem is $\min \operatorname{tr}(A^T X)$ s.t. $\operatorname{tr}(B^T X B) \geq \gamma$, $\operatorname{tr}(CX) \leq \delta$, $X \succeq 0$. Its objective and constraints are linear in X plus a PSD cone, so it is convex. I discussed the meaning of each constraint (a minimum controllability-type level and a bound on output variance) and the simplifications when A is symmetric and B is invertible. The report's solution satisfies $\operatorname{tr}(B^T X B) = 1.5 \geq \gamma$ and $\operatorname{tr}(CX) = 0.4982 \leq \delta$, with eigenvalues of $X^\star$ equal to 0 and 0.493. The script needs CVX, which was not installed here, so it was not re-run.